

Unit 2 Chemical Reactions

Writing Chemical Equations

Chemical equations allow us to write a short form description of a chemical reaction.

Like in mathematics, chemical equations have a structure that we must follow for them to make sense. In general, this form is:



A reactant is a chemical that is used up during a chemical reaction.

A product is a chemical that is created during a chemical reaction.

For each reactant and product, we should include the state of matter. We do this by including the state symbol as a subscript after compound:

Solid	(s)
Liquid	(l)
Gas	(g)
Aqueous	(aq)

*** If a substance is dissolved in water, it is said to be aqueous. ***

For Example:

Table Salt	$\text{NaCl}_{(s)}$
Salt dissolved Water	$\text{NaCl}_{(aq)}$

What would $\text{NaCl}_{(l)}$ be?

molten salt

General Equation Types

Spontaneous

- occurs without any special conditions / energy input
- is not necessarily a rapid reaction

Non-spontaneous

- requires a great deal of energy
- usually requires a *catalyst*

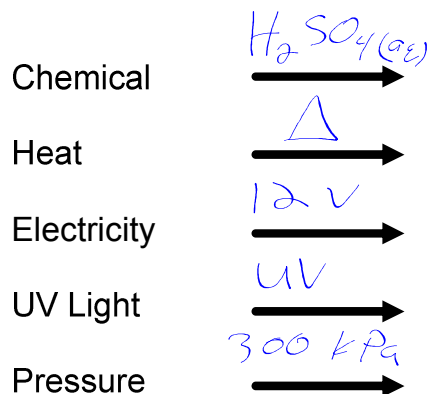
Exothermic

- heat is produced / given off

Endothermic

- heat is required / absorbed from the environment into the chemical reaction
- feels cold

A *catalyst* is a substance that speeds up / forces a chemical reaction. It is neither a reactant, nor a product - i.e. it is not a participant in the chemical reaction



Examples of word equations:

iron + oxygen $\longrightarrow$ iron(III) oxide



copper + silver nitrate $\longrightarrow$ silver + copper(II) nitrate



zinc + hydrochloric acid $\longrightarrow$ hydrogen + zinc chloride

